

Bayesian updating and reliability analysis for nuclear containment buildings

Donatien Rossat^{1,2}, J. Baroth¹, F. Dufour¹, M. Briffaut¹, B. Masson², A. Monteil², S. Michel-Ponnelle³

¹Laboratoire 3SR

²EDF/DIPNN/DT

³EDF/R&D/ERMES

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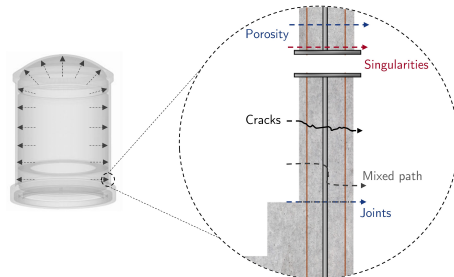
- Double-walled **Nuclear Containment Buildings (NCB)** are reinforced and prestressed concrete structures...
- ...whose leak tightness plays a key role in the safety of their surrounding environment
- Leak tightness of NCB is assessed through periodic **pressurization tests** aiming at measuring their leakage rate



Typical lifetime of a double-walled nuclear containment building

Introduction

- Leak tightness of NCB may evolve in time, under the action of several physical processes related to **concrete ageing**



Examples of leakage paths through a NCB wall during a pressurization test

- Necessity to develop a **probabilistic forecasting methodology** aiming at assessing the **risks** of exceeding a critical leakage rate threshold...
- ...whose predictions may be **updated** by assimilating **monitoring data** (delayed strains, leakage rate)

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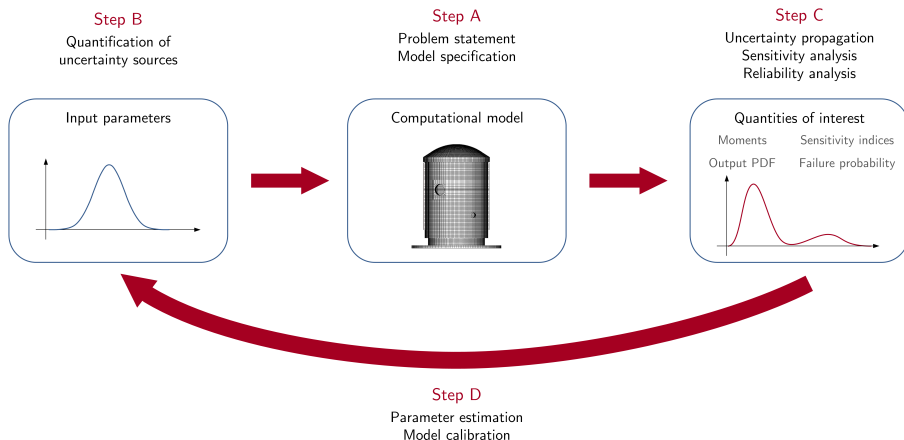
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General framework of Uncertainty Quantification

- General methodology based on the **Uncertainty Quantification (UQ)** framework:



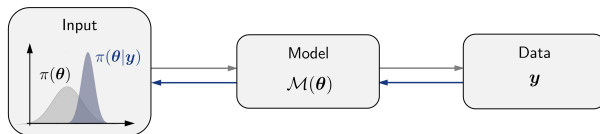
Bayesian inference for inverse problems

- Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space
- Let $\mathcal{M} : \mathcal{D}_\theta \subset \mathbb{R}^d \longrightarrow \mathcal{D}_y \subset \mathbb{R}^n$ a deterministic **model** with random **input parameters** $\Theta : \Omega \longrightarrow \mathcal{D}_\theta$
- The level of knowledge about Θ is summarized by a **prior** density $\pi(\theta)$
- Given observed data $y \in \mathcal{D}_y$, the **posterior** density of input parameters may be derived with Bayes' theorem:

$$\pi(\theta|y) = \frac{\pi(\theta)\mathcal{L}(\theta; y)}{\mathcal{Z}}$$

where $\mathcal{L}(\theta; y)$ is the **likelihood function**, and $\mathcal{Z}$ the **model evidence**:

$$\mathcal{Z} = \int_{\mathcal{D}_\theta} \mathcal{L}(\theta; y)\pi(\theta)d\theta$$



The BuS framework

- Bayesian computations are performed in the **BuS (Bayesian Updating with Structural Reliability methods)** framework, which reformulates Bayesian inference into a reliability problem (Straub & Papaioannou, 2015)

- Sampling from the posterior $\pi(\boldsymbol{\theta}|\mathbf{y})$ is tantamount to estimating the probability of the following failure event:

$$\mathcal{F} = \{(\boldsymbol{\theta}, p) \in \mathcal{D}_{\boldsymbol{\theta}} \times [0, 1] \mid p \leq c\mathcal{L}(\boldsymbol{\theta}; \mathbf{y})\}$$

where $c > 0$ is a constant chosen such that $c\mathcal{L}(\boldsymbol{\theta}; \mathbf{y}) \leq 1$, for all $\boldsymbol{\theta} \in \mathcal{D}_{\boldsymbol{\theta}}$

- Samples that lie in $\mathcal{F}$ are following the posterior distribution

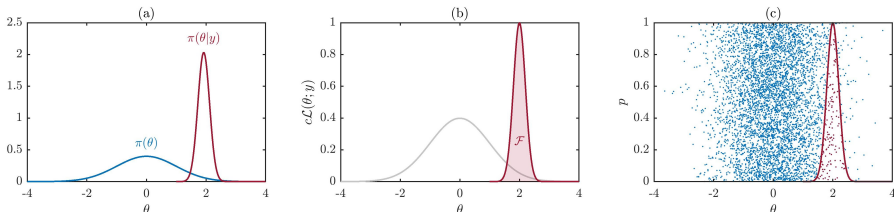


Illustration of the BUS framework: (a) Bayesian inference; (b) Equivalent failure event $\mathcal{F}$; (c) Solving the equivalent reliability problem through Monte Carlo Simulation

Bayesian computations with Subset Simulation

- The **Subset Simulation (SuS)** is a widely used technique for estimating probabilities of rare events:

$$\mathbb{P}_{\mathcal{F}} = \mathbb{P} \left(\bigcap_{i=0}^r \mathcal{F}_i \right) = \prod_{i=1}^r \mathbb{P}(\mathcal{F}_i | \mathcal{F}_{i-1})$$

where $\mathcal{F} = \mathcal{F}_r \subset \mathcal{F}_{r-1} \subset \dots \subset \mathcal{F}_1 \subset \mathcal{F}_0$, with $\mathbb{P}(\mathcal{F}_0) = 1$

(Au & Beck, 2001)

- ▶ Use a SuS algorithm adapted to the BuS framework
- ▶ Acceleration of Bayesian computations with **surrogate models**

(Betz et al., 2018)

(D.R. et al., 2022)

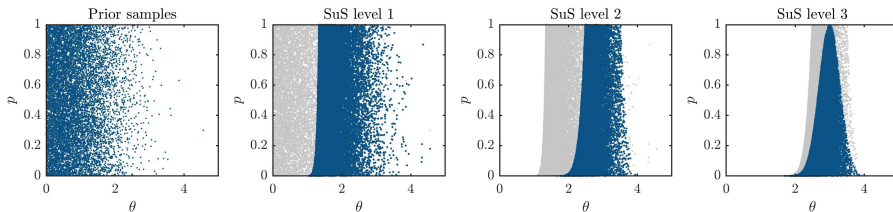


Illustration of SuS procedure, in the case $\pi(\theta) = \mathcal{N}(\theta|0, 1)$ and $\mathcal{L}(\theta; y) = \mathcal{N}(\theta|2, 0.2)$

Bayesian updating of failure probabilities

- Possibility to update failure probabilities in the BuS framework, given observation data (Straub et al., 2016)
- Typical quantity of interest in the framework of risk assessment for NCB:

$$\mathbb{P}(\mathcal{G}) = \mathbb{P}(\{\boldsymbol{\theta} \in \mathcal{D}_{\boldsymbol{\theta}} \mid Q(\boldsymbol{\theta}) > q_*\})$$

where Q is the leakage rate of the studied NCB, and q_* a regulatory leakage threshold

- Given observed data $\mathbf{y} \in \mathcal{D}_{\mathbf{y}}$, the updated failure probability $\mathbb{P}(\mathcal{G}|\mathbf{y})$ reads:

$$\mathbb{P}(\mathcal{G}|\mathbf{y}) = \frac{\mathbb{P}(\mathcal{F} \cap \mathcal{G})}{\mathbb{P}(\mathcal{F})}$$

where $\mathcal{F}$ is the BuS equivalent failure event associated to $\mathbf{y}$.

- Estimation of the probability $\mathbb{P}(\mathcal{G}|\mathbf{y})$ through a two-stage SuS algorithm

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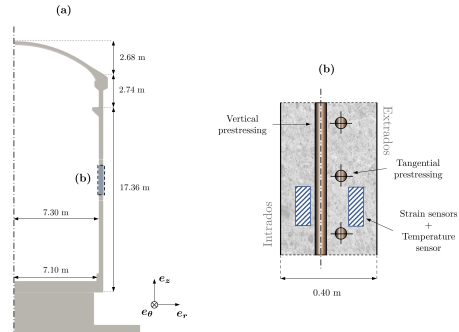
Conclusions

Application to NCB

- Application of the Bayesian updating methodology to nuclear containment buildings



Mock-up of a double-walled NCB, EDF-LAB Les Renardières

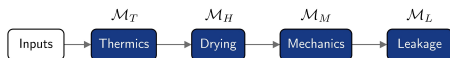


(a) 2D view of a typical NCB; (b) Focus on a standard zone

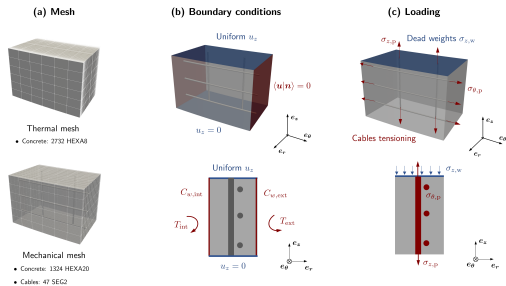
Modeling the physical behavior of NCB

- Modeling of the spatio-temporal evolution of thermal, hydrous, mechanical and transfert quantities of interest (e.g. water saturation, strains, leakage rate)
- Use of a **weakly-coupled thermo-hydro-mechanical & leakage (THML)** model: (Bouhjiti et al., 2020; D.R. et al., 2022)

$$\mathcal{M} = \mathcal{M}_L \circ \mathcal{M}_M \circ \mathcal{M}_H \circ \mathcal{M}_T$$



Weakly-coupled THML modeling



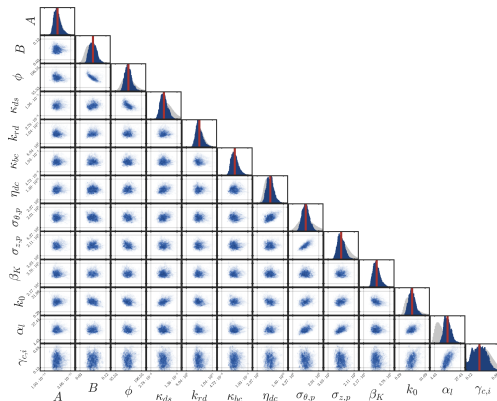
(a) Finite element mesh; (b) boundary conditions; (c) loading

Bayesian updating of model parameters

- Bayesian updating of model parameters using *in-situ* strain and leakage monitoring data

Updating of 13 primary influential input parameters

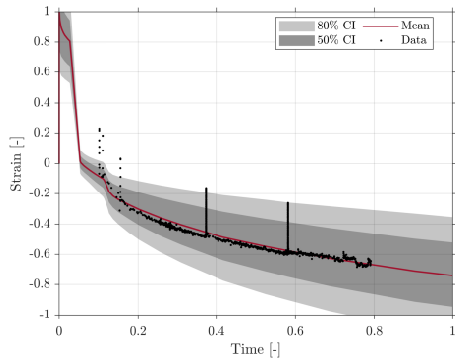
- ▶ 3 parameters related to the thermo-hydric behavior:
 A, B, ϕ
- ▶ 6 parameters related to the mechanical behavior:
 $\kappa_{ds}, k_{rd}, \kappa_{bc}, \eta_{dc}, \sigma_{\theta,p}, \sigma_{z,p}$
- ▶ 4 parameters related to the leakage behavior:
 $\beta_K, k_0, \alpha_l, \gamma_{c,i}$



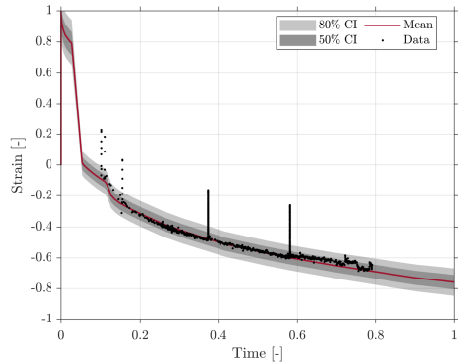
Posterior samples of input parameters of the THML model

Probabilistic delayed strains predictions

- Probabilistic predictions of the structure's long-term mechanical behavior
- Bayesian updating from *in-situ* measurements of delayed strains



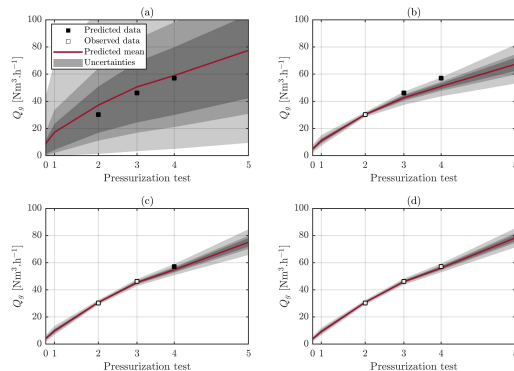
Prior tangential delayed strains of the standard zone



Posterior tangential delayed strains of the standard zone

Probabilistic global leakage predictions

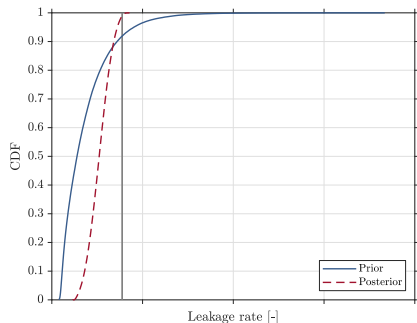
- Probabilistic predictions of the **global leakage rate** of the structure



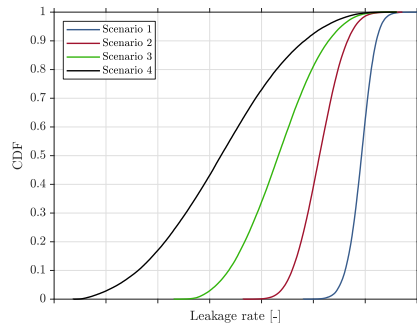
Example of Bayesian predictions of the global leakage rate: (a) prior; (b) posterior, Test 1; (c) posterior, Test 2; (d) posterior, Test 3

Reliability analysis and evaluation of the impact of repair works

- Estimation of the risk of exceeding a prescribed leakage threshold
- Evaluation of several repair works scenarios



Prior and posterior CDF of the structural leakage rate and comparison to a leakage threshold



Leakage rate CDF for several coatings scenarios

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- Uncertainty quantification methodology for Bayesian updating of predictions of the multi-physic behavior of nuclear containment buildings
- Use of a mathematical framework which allows to update both input parameters and failure probabilities
- Updating of uncertain input parameters and predictions of the structure's physical behavior throughout their operational phase
- Quantification of the risk of exceeding leakage regulatory thresholds, and evaluation of the impact of eventual repair works $\Rightarrow$ provides a decision aid tool for better planning maintenance operations

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